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Uncovering key success factors for enhanced office-site communication through digital twins

Azita Chalipour, Esmatullah Noorzai, Mahmood Golabchi and Seyed Hossein Hosseini Nourzad

Department of Project and Construction Management, School of Architecture, University of Tehran, Tehran, Iran

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Abstract

Purpose – This study aims to explore the transformative potential of digital twins (DTs) in the construction industry by addressing challenges and bridging the technology adoption gap. Focusing on construction offices, on-site environments and their interface, the research identifies key success factors for DT implementation. Emphasizing practical applications, the study foresees a substantial impact on project management, highlighting the pivotal role of DTs in reshaping the construction sector.

Design/methodology/approach – We reviewed 30 variables within the DT literature, classifying investigations into three main domains: on-site environments, office environments and site-office interface in construction projects. We employed a questionnaire to examine variables, with collected data analyzed using SmartPLS version 4 software. Confirmatory analysis and structural equation modeling rigorously assessed the model's reliability, validity and predictive utility.

Findings – The study identifies 11 critical factors crucial for successfully implementing DTs in construction projects, emphasizing contributions from office-related (ORF), site-related (SRF) and office-site-related factors (OSRF). Notably, impactful contributions from ORF affirm its pivotal role. The research sheds light on specific aspects within construction offices, on-site environments and their interface, providing valuable insights for industry stakeholders, policymakers and researchers.

Originality/value – Despite discussions on DT's potential, a research gap exists in addressing essential factors for its effective application in monitoring and optimizing construction projects throughout their lifecycle. This study fills this gap by identifying and categorizing these factors, enhancing the knowledge base for successful DT assimilation and emphasizing the need for practitioners to understand key factors crucial for its integration into construction processes.

Keywords Digital twin, Construction industry, Project lifecycle, Influencing factors, Project management, Structural equation modeling

Paper type Research paper

1. Introduction

Precision monitoring and control of assets throughout their lifecycle, spanning from manufactured components to infrastructure like buildings, has spurred significant exploration across various research domains into the possibilities and applications of digital twins. Technological advancements have played a pivotal role in driving this investigation forward (Boje *et al.*, 2020). However, the construction industry, a vital driver of economies globally, faces a dual challenge: the imperative for innovation to enhance efficiency and the lag in technology adoption compared to other sectors (Akanmu *et al.*, 2021; Tuhaise *et al.*, 2023). Despite contributing significantly, with 8–10% of GDP and fostering growth, job creation, and cross-industry synergy (Gharouni Jafari *et al.*, 2014; Noorzai *et al.*, 2017; Tuhaise *et al.*, 2023), it lags behind sectors like manufacturing and automotive in embracing technological advancements (Lin and Wu, 2021). For successful integration of technology in construction, it's vital to address industry challenges at their roots (Dadkani *et al.*, 2021). These challenges include low productivity, a negative industry image, unpredictability, and insufficient

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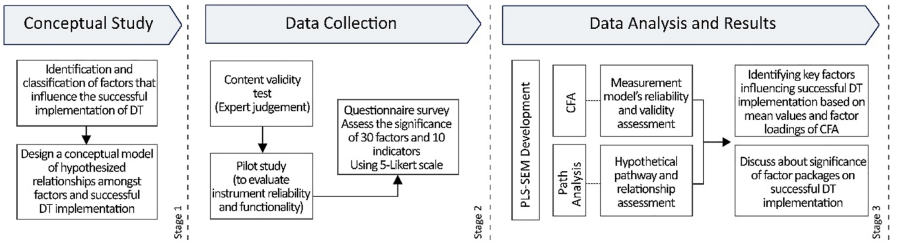
innovation investment (Ghorbany *et al.*, 2024; Mortezaei Farizhendy *et al.*, 2020; Farmer, 2016). The industry also grapples with inefficiencies in various aspects, including design, logistics, execution, and workforce management, underscoring the critical areas that demand attention (Hu *et al.*, 2022). Furthermore, the complexity and fragmentation of the construction process exacerbate data management challenges, affecting collaborative efforts, security, and technical information exchange (Zhang *et al.*, 2022a). Collaborative efforts suffer due to the industry's fragmented nature, necessitating streamlined communication for success (Lee *et al.*, 2021).

In this regard, the concept of the digital twin (DT) shows great promise in addressing these challenges effectively (Hu *et al.*, 2022). This holistic approach not only optimizes data management and coordination in construction projects but also enhances outcomes by seamlessly integrating advanced technologies across various facets of the industry (Noorzai, 2023; Zhang *et al.*, 2022a). Remarkably, a noteworthy majority of construction issues, ranging from 50% to 80%, stems from either data insufficiencies or delays in information acquisition, underscoring the pivotal role of accurate and timely information sharing in ensuring the cost and time efficiency of construction projects (Golabchi and Noorzai, 2013; Marzouk and Enaba, 2019). The DT concept, driven by advancements in Cyber-Physical Systems (CPS), the Internet of Things (IoT), sensor technology, and artificial intelligence (AI), establishes a robust physical-digital connection. This connection facilitates the immediate capture and transmission of real-time data to a digital platform (Ammar *et al.*, 2022). Subsequently, this data can be promptly disseminated among relevant project stakeholders, enabling them to actively monitor the project's current status, conduct comprehensive data analysis, and participate in closed-loop feedback processes that span the entirety of the construction project life cycle (Ammar *et al.*, 2022; Lee *et al.*, 2021). This collaborative approach is essential for ensuring transparent and accountable information sharing among the various stakeholders involved in the digital twin ecosystem (Lee *et al.*, 2021). Moreover, digital twin solutions have the capacity to monitor and manage additional resources like materials, labor, and equipment, enhance worker safety through virtual site simulations, and offer predictive capabilities. In construction project management, digital twin usage entails integrating sensor data into its database to rigorously compare planned ideals with actual progress (Attaran and Celik, 2023). These Twins also serve as a centralized repository for project information, streamlining processes and providing clarity on project timelines (Ammar *et al.*, 2022). This enables precise tracking, generates crucial reports, and provides essential alerts for effective project management oversight (Attaran and Celik, 2023).

In the contemporary construction landscape, fraught with challenges such as inefficiencies and delays, the emergence of DTs stands out as a transformative solution (Golabchi *et al.*, 2016). This study addresses a critical void in the existing literature by unraveling key factors crucial to successfully implementing DT in monitoring and controlling construction projects across their various phases. This underscores the need for a nuanced understanding of DTs in construction and highlights the importance of overcoming existing limitations for more effective DT implementation. To address the research deficiency, this study focuses on the collaborative interaction between DT technology and the various stakeholders involved in construction projects. This investigation is categorized into three key areas: "DT in Construction Site Environments", "DT in Construction Office Environments", and "DT in Construction Site-Office Interface". Within each of these domains, the authors delve into the factors that must be taken into account to effectively use DT technology for monitoring and controlling construction projects throughout their life cycles. The analysis in this research also seeks to determine the significance of these identified factors within each domain, ultimately leading to a comprehensive assessment of how they collectively impact the successful adoption of DT technology.

As illustrated in Figure 1, the following section of this paper discusses the theoretical basis for conceptualizing DT technology in construction projects across their life cycle. The development of hypotheses and the creation of a conceptual model are then outlined.

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Figure 1. Methodology employed in this study

This model is validated through empirical data from surveys using structural equation modeling (SEM). Confirmatory Factor Analysis (CFA) is employed to factor classification and to test the hypothesized relations among these factors within the three packages. SEM is also utilized to elucidate the relationships and pathways connecting these three packages with the successful implementation of DT. The findings related to these objectives are discussed to highlight their contributions to knowledge and practical applications. The article concludes by addressing limitations and offering recommendations for future research. In essence, this research aims to bridge the gap between theoretical concepts and practical applications within the construction industry. By focusing on critical domains of construction projects, actionable insights are provided to improve communication, collaboration, and project efficiency. The primary objective of this research is to offer construction professionals a practical guide for effective DT technology implementation, meeting the evolving demands of the sector and promoting smoother, more efficient construction processes.

2. Literature review

2.1 Concept and definitions of DT

The concept of a "digital twin" originated in the aerospace sector, particularly during NASA's Apollo project in the 1960s. The project involved creating two space vehicles, with one, known as "the twin," on Earth, faithfully replicating the actions of the space vehicle on the mission in real-time, serving as a prototype mirroring its operational activities (Boschert and Rosen, 2016; Madubuike *et al*., 2022). Subsequently, the term "digital twin" was coined by Michael Grieves in 2003 during his course on product lifecycle management, marking the inception of the widely acknowledged term "digital twin" (Grieves and Vickers, 2017). In 2010, Michael Grieves coined the term "digital twin" after exploring the "Information Mirroring Model" in 2006 (Grieves, 2014). This concept transitioned from a product lifecycle management tool to a comprehensive digital platform (Tao *et al*., 2018). NASA formally defined DT in 2012 as an integrated, multi-physics, multi-scale simulation of a system replicating real-world operations using advanced models, sensor data, and historical information (Agrawal *et al*., 2023). This evolution yielded two key features: (1) integrated product lifecycle support and (2) real-time data generation and management through cognitive technologies (Ozturk, 2021).

DT characterization extends beyond industry-specific boundaries, focusing on the evolving behavior of physical entities throughout their lifecycle (Opoku *et al*., 2021). Various authors have proposed diverse definitions. In 2015, (Rosen *et al*., 2015) defined DT as a fusion of physical and virtual spaces for assessing the lifecycle operations of a physical component. A more concise definition by Boschert and Rosen (2016) emphasized DT's role in gathering relevant data, particularly data exchange and control algorithms. DT can be seen as a digital representation of a physical object, as described by Canedo (2016). (Grieves and Vickers, 2017) highlighted DT's role in product lifecycle management, involving virtual constructs that

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represent a product's components and structure, enhancing quality and reducing design-execution disparities, offering more detailed insights compared to Grieve's 2006 definition, particularly within the realm of product lifecycle management. (Liu *et al*., 2018; Madni *et al*., 2019) highlighted DT's capacity for real-time data collection from physical assets and its significance in construction, enabling real-time monitoring and predictive maintenance, all driven by predictive models based on accumulated data. (Kor *et al*., 2023) underscores DT's role in controlling CPSs, while (Tuhaise *et al*., 2023) highlights its adaptability in updating BIM and real-time asset monitoring, both recognizing its versatility and value in enhancing decision-making across various contexts (Asgari and Noorzai, 2021; Ghorbany *et al*., 2023). DT is broadly defined by researchers across various industries. Table 1 presents a chronological list of DT definitions, along with references and their applicability across diverse fields and lifecycle stages.

2.2 Technologies for applying DT

DTs record various aspects of construction projects, including changes, timelines, reporting sources, issues, resolutions, and responsible parties (Struková *et al*., 2022). DTs rely on key technologies to gather real-time data, extract insights, and create digital representations of physical objects, as revealed in research findings (Attaran and Celik, 2023). These technologies include:

Internet of Things (IoT): IoT connects entities, collects real-world data through sensors, and continuously updates digital object replicas, serving as a core technology in all DT applications.

Cloud Computing: Cloud services offer internet-based storage and data processing, overcoming storage limitations and accelerating data processing for complex systems.

Artificial Intelligence (AI): AI, including machine learning and deep learning, automates data analysis, provides insights, predicts outcomes, and enhances decision-making.

Extended Reality (XR): XR encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), merges physical and virtual worlds to create interactive real-time models, facilitating user engagement and understanding.

2.3 Workflow

In general, creating a digital twin involves establishing a data link between a physical object and its virtual counterpart through modeling technologies that replicate the physical entity's geometric characteristics, operational capabilities, current condition, spatial coordinates, operational procedures, and overall performance (Wang *et al*., 2022).

IoT technologies establish bidirectional data exchange between physical and virtual entities (Golabchi *et al*., 2016). Data is collected from the physical environment using sensors and transmitted to the virtual model via application layer protocols. Big data storage solutions are essential for managing voluminous and diverse data (Agrawal *et al*., 2023). Advanced analytics processes digital twin data, providing various user services (Kor *et al*., 2023). As illustrated in Figure 2(a.), Advanced data analytics, in conjunction with actuators to interact with the physical environment (Radanliev *et al*., 2022), can be leveraged for processing digital twin data, thereby providing a diverse array of user services. The processed data becomes readily accessible to end users through interactive data visualization supported by visualization technologies (Qi and Tao, 2018). These technologies enable a functional digital twin with a high-fidelity model capable of bidirectional data transfer and data processing (Tuhaise *et al*., 2023).

It is noteworthy to position the workflow framework on the DT maturity spectrum, as outlined in (Pregnolato *et al*., 2022), which outlines five actionable steps for a systematic and integrated development of DT components.

(1) Data gathering: collecting real-world information for model support.

(2) Digital modeling: creating virtual models for simulating and controlling structures.

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Table 1. DT definitions in the literature

No	Reference	Year	Definition	Keywords	Application	Phase
1	Shafto *et al.* (2010)	2010–2012	The term "digital twin" refers to an integrative multi-physics, multi-scale, probabilistic simulation of a vehicle or system that employs the finest physical models currently available, sensor updates, fleet history, etc. To replicate its flying twin's behavior	integrative	NASA's integrated simulations	Life cycle
2	Lee *et al.* (2013)	2013	A digital twin is a connected replica of a genuine machine that is tied together, runs on a cloud platform, and simulates its state of health using information from both data-driven analytical algorithms and other physical knowledge that is readily available	connected replica	Predictive manufacturing	Life cycle
3	Grieves (2014)	2014	A Digital Twin is a unified virtual representation of an asset, process, or system that mirrors the behavior and performance of its physical counterpart, providing valuable insights, comprehension, and enhanced visibility into intricate processes	unified virtual representation	General	Life cycle
4	Rosen *et al.* (2015)	2015	Digital twins are highly accurate representations of both the present condition of a process and its behaviors as they interact with their real-world environments	highly accurate representation	Industrial manufacturing	Life cycle
5	Gabor *et al.* (2016)	2016	A digital representation of the actual physical object that enables future system state forecasting	future system state forecasting	System design	Life cycle

(*continued*)

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Table 1. Continued

No	Reference	Year	Definition	Keywords	Application	Phase
6	Grieves and Vickers (2017)	2017	A group of virtual information constructions that completely define a produced product, whether it be potential or real, down to the atomic level and up to the large-scale geometrical level	virtual information constructions	Product lifecycle management	Life cycle
7	Zhuang *et al.* (2018)	2018	A virtual, dynamic model in the digital realm mirrors a real-world entity precisely, simulating its characteristics, behavior, lifespan, and performance in real-time	dynamic model in the digital realm	Smart manufacturing	Life cycle
8	Madni *et al.* (2019)	2019	A virtual representation (twin) of a physical system, continuously updated with real-data on the physical system's performance, maintenance, and health throughout its lifecycle	virtual representation, continuously updated	General	Life cycle
9	Hu *et al.* (2021)	2020	a virtual counterpart of physical entities, including assets, processes, individuals, locations, and systems. It captures both their components and dynamic behavior as they evolve throughout their lifecycle	capturing both the components and dynamic behavior	General	Life cycle
10	Madubuike *et al.* (2022)	2021	a virtual replica of a physical asset, generated using technologies like sensors, communication networks, and 3D models. It facilitates real-time updates and bi-directional coordination, ensuring an accurate virtual replica of the physical asset	real-time updates, bi-directional coordination	Construction	Life cycle

(*continued*)

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Table 1. Continued

No	Reference	Year	Definition	Keywords	Application	Phase
11	Opoku *et al.* (2023)	2022	Real-time depiction of a completely or partially finished structure or building that reflects the traits and stage of completion	Real-time depiction	Construction	Life cycle
12	Tuhaise *et al.* (2023)	2023	The digital twin has arisen as a method for updating a BIM model with current data to achieve cyber-physical integration, allowing for real-time monitoring of assets and operations and enhancing decision-making	real-time monitoring	Construction	Life cycle

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(3) Sensor data collection: efficiently gathering sensor data while complying with standards.

(4) Data/model integration: facilitating connections between real and virtual components.

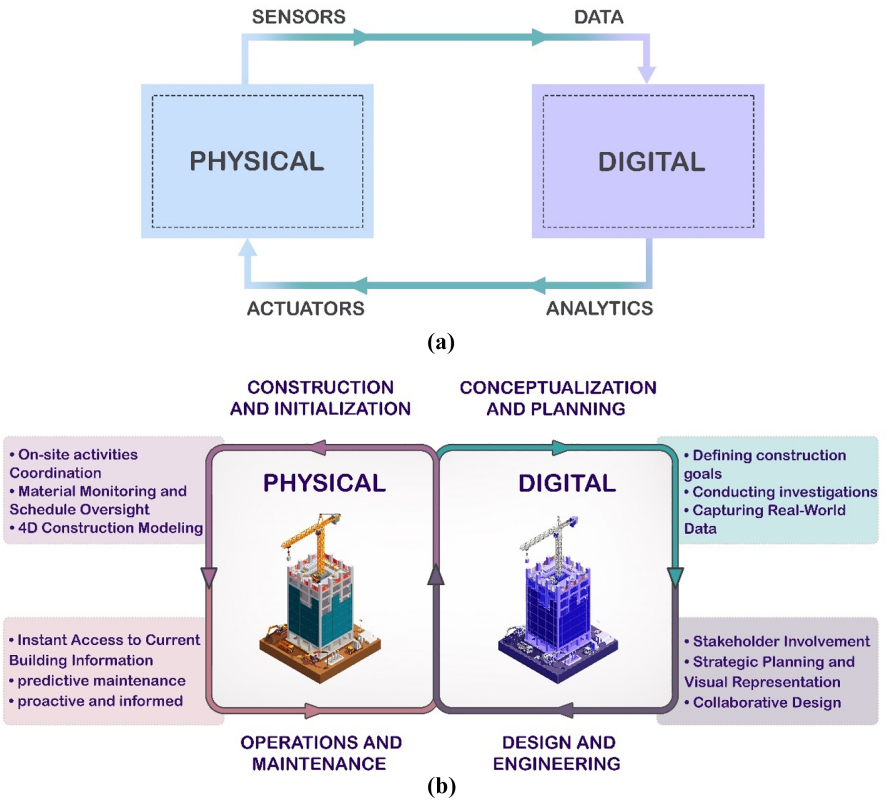
(5) Operational support: enhancing performance and autonomy through DT integration in decision-making.

2.4 Background and related works

2.4.1 The use of DT in construction. In recent years, rapid advancements in DT technology have significantly transformed information creation, storage, and sharing in the construction industry. These transformations are primarily attributed to the widespread use of BIM and the integration of DT with technologies like the IoT and AI (Xie *et al*., 2022). DTs, unlike BIM, stand out for their capacity to incorporate real-time cyber-physical interactions, enabling seamless real-time integration and interaction between the physical environment and its digital representation Alsuhaibani *et al*. (2022). This facilitates continuous progress monitoring and dynamic control (Jiang *et al*., 2022). In this context, (Delgado and Oyedele, 2021) explored the distinction between BIM and DT, emphasizing their varying roles tailored to industry-specific needs. (Boje *et al*., 2020) also highlight that while BIM focuses on procedures, technologies, and data structures, DT offers a broader socio-technical perspective through real-time data flow between physical and digital realms. They also point out BIM's limitations in integrating control systems and sensor networks for dynamic data exchange.

Several studies have explored the multifaceted applications of DT in the construction industry. For instance, (Zhang *et al*., 2022a) proposed a framework to enhance BIM through DTs for construction site management. It improves DT mechanisms, including BIM-based digital representation, IoT integration, data management, analytics, and interaction with the physical environment. This framework aims to optimize construction site processes, including incident response, work efficiency, emissions, safety, quality, and administrative aspects. Following this, (Pan and Zhang, 2021) introduced a closed-loop digital twin framework incorporating data mining (DM), the Internet of Things (IoT), and BIM approaches with a

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Figure 2. (a) Digital twin workflow. (b) Digital twins throughout lifecycle phases

focus on smart construction project management. IoT gathers real-time information bridging the digital and physical worlds, while DM uncovers hidden insights in this data. A BIM-based project that used IoT data to create a 4D visualization model and task-centered process simulations served as a practical validation of this framework. Moreover, (Lin and Wu, 2021) contribute to this growing body of research by introducing a novel approach for twinning and mining collaborative networks, substantiated by empirical data gleaned from on-site inspections during a construction project. This method presents a strategic framework aimed at enhancing existing information systems by including automated data collection and network formation elements, hence addressing the widespread manual processes involved in collaborative network construction. In a related context, Greif *et al*. (2020) introduced a digital twin framework to optimize the digital supply chain by creating digital replicas of construction material silos and obtaining real-time fill-level data through sensor and IoT integration. This model allowed real-time monitoring and enhancement of silo transportation to building sites, yielding substantial time and cost efficiencies in material logistics and replenishment planning. These studies underscore the necessity of addressing fundamental challenges and their underlying causes to improve the reliability and adoption of DTs in the construction sector.

2.4.2 DT in Construction Office Environments. The concept of digital twins is a valuable tool for improving management processes in the construction industry. By utilizing artificial intelligence and computer-aided engineering, DTs enhance performance, reduce costs,

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mitigate risks, and optimize supply chains (Salem and Dragomir, 2022). They play a crucial role in enhancing user interaction in smart construction projects (Agrawal *et al*., 2023), particularly in facilitating communication and collaboration between on-site and office-based teams. Through real-time 3D BIM designs, DTs monitor construction sites, report potential issues, and provide data analytics by aggregating information from various sources (Sepasgozar *et al*., 2020). This enables real-time access for on-site and office teams to monitor equipment, machinery, and assets, addressing needs, anticipating challenges, and facilitating autonomous solutions. The constant updates on physical asset status and location promote data-driven collaboration, significantly enhancing project execution (Ghansah and Lu, 2023). DTs serve as a bridge between efficient user interaction, enhanced planning, and seamless project execution within the construction industry. For instance, (Hasan and Sacks, 2023) introduced a comprehensive system for managing production information across the building lifecycle. A Digital Twin Construction (DTC) system was implemented to provide seamless data transfer and monitoring between the field and BIM, thereby facilitating automated, accurate, and intelligent production control. The system assists planners in optimizing lookahead planning (LAP) and refining decision-making procedures. Additionally, the authors provide an event processing model (EPM) that employs numerous monitoring methods concurrently, with algorithms to analyze data from the multiple diverse streams, to enhance operational control. Moreover, (Piras *et al*., 2024) presented a digital management approach to mitigate the delays occurring during construction projects by integrating DT, BIM, IoT, and Artificial Intelligence (AI). It basically focuses on optimizing project management through real-time data flow and monitoring, enhancing scheduling, decision-making, and resource allocation. The methodology employs 4D BIM models and IoT-enabled monitoring to facilitate predictive analysis, improve coordination, strengthen operational control, expedite management procedures, and minimize delays, so providing a more innovative and sustainable solution for construction management. Notably, there is a research gap regarding DT applications in office environments of construction projects and the essential considerations for successful DT implementation.

2.4.3 DT in construction site environments. Digital Twins are revolutionizing construction sites by enabling data integration between machines, workers, and processes, fostering informed decisions through IoT and sensors (Kosse *et al*., 2022). They enhance real-time transparency, track resources, and access to historical data to tackle construction's inherent unpredictability. Furthermore, DTs provide automated navigation, alerts, and real-time schedule adjustments for on-site workers, fostering efficient cooperation and enhancing construction site efficiency and control (Jiang *et al*., 2022). They also play a vital role in ensuring construction site safety by monitoring worker presence, safety compliance, and detecting abnormal behaviors and fatigue in real-time, significantly improving overall site safety. Additionally, DTs adeptly identify potential conflicts in project plans through 4D clash simulations, addressing data reliability issues and allowing construction teams to examine various scenarios for superior project management (Boje *et al*., 2020). For instance, Jiang *et al*. (2022) introduced DT-SYNC, a real-time synchronization system powered by DTs for optimizing on-site precast assembly, particularly in urban environments with spatial constraints. It offers a smart construction site framework leveraging DTs for real-time data, aiding informed decision-making during construction. DT-SYNC efficiently coordinates on-site tasks and resource allocation, ensuring robust construction management, even in uncertain conditions. Additionally, Alizadehsalehi and Yitmen (2023) presented a framework for an automated construction progress monitoring system based on DT technology. This framework integrates BIM, DTs and XR technologies (VR, AR, and MR), outlining steps on how these technologies work together to create, capture, produce, and analyze construction data. The research findings confirm the positive impact and importance of technology integration within a coherent framework, which provides reliable, real-time, transparent, and digital construction progress monitoring. In conclusion, DTs offer a comprehensive solution for enhancing construction site dynamics, safety, clash detection, and logistics. It is noteworthy to

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emphasize that this research is dedicated to exploring pivotal factors crucial for the successful implementation of DTs in the site environment.

2.4.4 DT in construction site-office interface. Digital Twins play a crucial role in connecting construction sites with office interfaces by providing real-time replicas of physical sites, enabling continuous monitoring, advanced simulations, and data-driven analysis. This integration greatly improves construction project management efficiency throughout its lifecycle, as per the findings in (Zhang *et al*., 2022a):

(1) Site Monitoring: Enables instant data analysis and emergency response through a real-time interface for managers.

(2) Equipment Management: Provides early warnings through visual equipment status.

(3) Asset Management: Instant access to asset data for informed decisions.

(4) Supply Chain Management: Streamlines logistics and resource use via supply chain simulations.

(5) Worker-Power Management: Supports resource allocation and cost management by analyzing workload data.

(6) Safety Analysis: Promotes accident prevention through structural integrity checks and personnel tracking.

(7) Emissions and Energy Management: Enhances performance via data from sensors on emissions and energy.

2.4.5 Applying digital twins throughout lifecycle phases. DT technology has advanced in manufacturing, aerospace, and computing, and is now entering the construction industry, serving the entire project lifecycle, including planning, design, construction, inspection, and maintenance (Xie *et al*., 2022), as illustrated in Figure 2(b) (Züst *et al*., 2021) highlight Digital Twin Construction (DTC) as a cutting-edge method in construction management, integrating data from site monitoring tech and AI. This fusion offers real-time data and proactive analysis, optimizing design, planning, and production. DTC ensures dynamic data flow and fosters ongoing knowledge enhancement for intelligent project lifecycle management.

DTs streamline planning and design by merging digital information with 3D product models, creating a strong data foundation for the project's lifecycle. Using DT in the design phase minimizes confusion and costly change orders during installation by meticulously recording project information (Xie *et al*., 2022). Employing an adaptable DT efficiently allows designers to assess alternatives, detect design imperfections, and access project data for informed decision-making and comprehensive tracking (Guo *et al*., 2019; Mykoniatis and Harris, 2021; Zhang *et al*., 2020). This collected data supports future projects by aiding material selection, energy management, procurement, and supplier choices, enabling stakeholders to enhance planning, clarify intentions, and manage project life cycle stages effectively (Opoku *et al*., 2021).

During the construction phase, often overlooked in DT technology literature, the focus shifts to real-time data collection, workforce allocation, and material scheduling at the site. This entails gathering information on labor, materials, machinery, project management, and planning, facilitated by IoT-based construction status reporting. Successful deployment of DTs in construction requires a robust model framework and specific implementation details (Xie *et al*., 2022). For instance, Pal *et al*. (2023) presented the concept of Digital Twin Construction (DTC), which integrates BIM, lean techniques, on-site data acquisition, and AI-driven analytics, and examined the application of DTs in construction to enhance progress monitoring and comprehensive project management. Their research is on the automation of construction progress monitoring (CPM) through using vision-based approaches and rectifying deficiencies in existing procedures and it also highlights the potential of DT to enable

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real-time, closed-loop project control, eventually improving productivity, efficiency, and decision-making in construction projects by examining several digital technologies, such as computer vision and machine learning. DTs wield substantial data resources in the construction phase, yielding cost savings, quality improvements, and enhanced stakeholder management by furnishing stakeholders with comprehensive project insights (Opoku *et al*., 2021).

During the operation and maintenance phase, Digital Twin applications support tasks like daily usage, fault repairs, maintenance, and potential demolition. Real-time data on service life and operation status enables automated analysis, predictive maintenance, and design improvements for future projects (Xie *et al*., 2022). Lu *et al*. (2020) presented a semi-automated geometric digital twin method that integrates images and CAD drawings for existing buildings in their operational and maintenance phases. This approach ensures immediate access to current building data, preventing delays in decision-making due to outdated information, and ultimately boosting organizational efficiency.

In construction, DT technology has been underutilized in the demolition and recovery phases. Despite the potential for valuable behavioral insights during this stage, maintaining a low-cost virtual representation is viable due to abundant lifecycle data (Liu *et al*., 2021). This technology is especially useful in conserving heritage assets facing potential demolition. Various studies have leveraged Historic Building Information Modeling (HBIM) to improve collaborative information management for existing structures (Opoku *et al*., 2021). Barazzetti *et al*. (2015) introduced an AR/VR method to create detailed HBIM for cultural tourism engagement. This approach integrates DT and HBIM processes, aiding building managers in identifying potential hazards, implementing technical solutions, and preserving assets (Jouan and Hallot, 2020). Notably, this review found no studies applying DT technology to the demolition and recovery phases of construction projects.

2.5 Research gap

In order to facilitate the successful implementation of DT technology for the monitoring and optimization of diverse construction processes, it is imperative to equip practitioners with a thorough understanding of the essential aspects and factors that ensure its effective integration across the entire lifecycle of construction projects. Regrettably, the existing body of literature falls short in adequately addressing these critical concerns. Despite the extensive discussion by researchers and industry practitioners on DT's potential in enhancing construction monitoring and optimization, a notable research gap persists within the academic domain. This gap highlights the absence of comprehensive investigations into pivotal factors for successful DT application in monitoring and controlling construction projects across diverse lifecycle phases. This study addresses the knowledge gap by identifying essential factors for effectively using DT in overseeing construction projects across diverse life cycle phases. These factors are categorized into three groups, as outlined earlier, for in-depth analysis within each domain, facilitating successful technology assimilation.

2.6 Systematic review and development of conceptual model

To conduct a comprehensive review within the established scope, this study initially relied on Scopus for literature searches due to its broad coverage of scientific (Opoku *et al*., 2021). As the number of relevant publications retrieved was limited, additional searches were performed on ScienceDirect, Elsevier, ASCE, Emerald, Springer, and Web of Science to ensure a sufficient number of relevant studies on the application of Digital Twins in the construction industry. The search terms were based on "article title/abstract/keyword" and split into three parts: the first related to "digital twin" or "digital replica"; the second to "construction" or "construction industry/site/field"; and the third to "project lifecycle," "project management," or "project offices." The publication window was set from 2020 to 2023, with a focus on articles and reviews, as these are generally regarded as the most credible sources of knowledge.

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Initially, a total of 114 journal and conference papers were identified. A preliminary screening was conducted to exclude studies that were not directly related to the construction industry or included keywords without substantive relevance. This process underlined that at least one relevant paper each was available in journals like Automation in Construction, Journal of Building Engineering, Journal of Management in Engineering, Journal of Architectural Engineering, Buildings, Sustainability, and Journal of Construction Engineering and Management. In the sequel to this stage, a total of 78 papers from 39 journals were retained, as represented by Table 2.

Table 2. Overview of selected publications relevant to the study

N.	Publication title	Year 2021	2022	2023	Total	2020
1	Advanced Engineering Informatics		1			1
2	Advances in Civil Engineering	1		1		2
3	Applied Sciences			1		1
4	ASCE International Conference on Computing in Civil Engineering		1			1
5	Assembly Automation		1			1
6	Automation in Construction	2	3	9	2	16
7	Buildings			5		5
8	Built Environment			1		1
9	Computers and Electrical Engineering			1		1
10	Computers in Industry	1		3		4
11	Construction Innovation			1		1
12	Construction Research Congress			3		3
13	Decision Analytics Journal				1	1
14	Digital Transformation and Society			1		1
15	Engineering, Construction and Architectural Management			2		2
16	Frontiers in built environment			3		3
17	IEEE Access			1		1
18	Informatic				1	1
19	International Conference on Construction and Real Estate Management			1		1
20	International Journal of Construction Management		1			1
21	International Journal of Human–Computer Interaction			1		1
22	International Journal of Intelligent Robotics and Applications			1		1
23	IOP Conference Publishing			2		2
24	Journal of Architectural Engineering			1		1
25	Journal of Asian Architecture and Building Engineering		1			1
26	Journal of Building Engineering		2	2	1	5
27	Journal of Cleaner Production				1	1
28	Journal of Computing in Civil Engineering				1	1
29	Journal of Construction Engineering and Management			2		2
30	Journal of Industrial Information Integration			2		2
31	Journal of Information Technology in Construction		1	1		2
32	Journal of Intelligent Manufacturing		1			1
33	Journal of Intelligent Manufacturing and Special Equipment		1			1
34	Journal of Management in Engineering			3		3
35	Management Decision			1		1
36	Smart and Sustainable Built Environment		1		1	2
37	Structure and Infrastructure Engineering		1			1
38	Sustainability	1				1
39	Technology Analysis and Strategic Management			1		1
	Total	*5*	*15*	*50*	*8*	*78*

Source(s): Authors' own work

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The second step was a critical and thorough review of the 78 identified papers for relevance on the research topic. Through reviewing research patterns, identifying gaps, and analyzing trends, this study narrowed the focus to three areas: "DT in Construction Site Environments," "DT in Construction Office Environments," and "DT in the Site-Office Interface." This was then followed by the identification of 15 key papers related to providing enough information on factors that are necessary for the successful implementation of DT in monitoring and controlling construction projects throughout their lifecycle with a focus on the collaborative interaction between DT technology and the various stakeholders involved in construction projects. Table 3 presents the final selection of the key papers. After a thorough literature analysis on successful DT implementation indicators, a concise conceptual model has been developed, outlining essential factors for effective DT application in monitoring and managing construction projects across various life cycle stages. These factors are grouped into "DT in Construction Site Environments", "DT in Construction Office Environments", and "DTs in Construction Site-Office Interface," referred to as "site-related", "office-related", and "office-site related" for simplicity. Based on insights from prior research, each group consists of ten distinct factors, totaling 30 factors presented in Table 3. In addition, 10 specific metrics for assessing DTs in construction projects have been compiled based on collaborative discussions. These metrics form the basis for this study's hypotheses and will undergo a rigorous examination to assess assumptions about DT implementation. This research explores potential connections between indicator packages and individual indicators, leading to the formulation of four hypotheses crucial for successful DT implementation in monitoring and controlling construction projects.

H1. All packages contribute to successful DT implementation of construction projects. The contributions and paths for improvement of these packages are different, but they all influence the successful DT implementation.

H2. The office-related package has a significant effect on the successful DT implementation.

H3. The site-related package has a significant effect on the successful DT implementation.

H4. The office-site related package has a significant effect on the successful DT implementation.

3. Research setting

This study was conducted in the context of the world construction industry, with a focus on specific DT applications that are being used to monitor and control construction projects. The research setting includes different geographic regions: Iran, North America, Europe, the Asia-Pacific, and the Middle East, as represented by the respondents participating in the survey. These regions were selected based on their different levels of advancement in DT and BIM technologies and their differing regulatory environments and market dynamics. Several studies highlight the advancements in these regions. For example, North America has focused on AI-driven DT applications for project optimization and Europe emphasizes sustainability-focused DT integration (Ammar *et al*., 2022; Kineber *et al*., 2023). The Asia-Pacific and Middle East regions are experiencing growth in DT adoption due to large-scale infrastructure projects and regulatory support (Attaran and Celik, 2023; Zhang *et al*., 2022a).

The target population for the survey involved professionals who possess considerable experience in DT implementation for construction projects, including project managers, BIM experts, project engineers, and academic researchers. The study also emphasized the collaborative roles played by clients, consultants, and contractors who influence DT adoption along the various phases of a project's life cycle. In addition, the research captures perspectives

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Table 3. Measurement model evaluation

Constructs	Measurement item	Reference	Code	Mean	Standard deviation	Rank	Loadings	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
Successful DT implementation [measured by indicators of DT implementation (DTI)]	Improved decision-making based on real-time data-driven insights	–	DTI1	3.948	0.934	8	0.682	0.894	0.913	0.513
	Improved performance management and more efficient workflow management	–	DTI2	3.889	0.861	10	0.689	–	–	–
	Real-time data access and sharing across all stakeholders	–	DTI3	4.173	0.872	2	0.788	–	–	–
	Centralized communication and collaboration platform and better integration of project information and data from multiple sources	–	DTI4	4.056	0.873	5	0.738	–	–	–
	Better visualization and understanding of project status and progress	–	DTI5	4.185	0.944	1	0.73	–	–	–
	Improved project planning and scheduling	–	DTI6	4.123	0.88	4	0.669	–	–	–
	Efficient monitoring and control of project activities and better reporting	–	DTI7	4.148	0.803	3	0.711	–	–	–
	Efficient risk management, minimized errors and rework, and enhanced change management processes	–	DTI8	4.049	0.942	6	0.705	–	–	–
	More efficient facility management and maintenance of the built environment	–	DTI9	3.938	0.947	9	0.697	–	–	–
	Increased efficiency and productivity in project delivery	–	DTI10	4.037	0.936	7	0.747	–	–	–
								–	–	–

(*continued*)

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Table 3. Continued

Constructs	Measurement item	Reference	Code	Mean	Standard deviation	Rank	Loadings	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
Office-related factors (ORF)	Integrating external factors with historical and current firm performance data to assess obstacles, opportunities, and potential outcomes for diverse strategic paths	2-10	ORF1	3.679	0.914	9	0.736	0.891	0.911	0.506
	Capturing functional needs through design parameters, working principles, and physical structures in the conceptual design stage	3	ORF2	3.951	0.915	3	0.639	–	–	–
	Offering an AI-powered analytical tool for automated data analysis, insights, predictions, and problem avoidance suggestions	9	ORF3	3.877	0.986	4	0.701	–	–	–
	conducting end-user requirements extraction and analysis in the digital twin design stage	5	ORF4	3.741	0.94	8	0.737	–	–	–
	Ensuring accurate status information and proactively optimizing ongoing design and planning	1-7	ORF5	3.619	0.853	10	0.698	–	–	–
	Supplying planners with current information and knowledge about the as-built product and as-performed processes as a basis for planning, reducing reliance on expert experience and potential misjudgment	12	ORF6	3.815	0.944	5	0.703	–	–	–
	Enable on-site inspection by providing all stakeholders with real-time collaborative access to updated information	10	ORF7	3.716	1.068	7	0.739	–	–	–
	Using project execution data for predictive what-if scenario testing to ensure a successful completion	2-10	ORF8	3.963	0.793	2	0.659	–	–	–
	Using real-time AI to proactively anticipate and offer actionable options for managers, or automate their implementation based on the DT database	15	ORF9	3.716	1.033	6	0.781	–	–	–
	Monitoring project status during execution to ensure compliance with contract documents and norms, along with automated inventory management for real-time updates	8-4-10	ORF10	4.012	1.048	1	0.732	–	–	–

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Table 3. Continued

Constructs	Measurement item	Reference	Code	Mean	Standard deviation	Rank	Loadings	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
Site-related factors (SRF)	Using sensing devices to enable specific functions in various construction site components, including space, boundaries, roads, facilities, equipment, materials, and personnel	11	SRF1	3.79	0.939	6	0.771	0.901	0.919	0.531
	Ensuring operator safety by providing effective on-site instructions using a virtual environment tailored to specific working conditions	6	SRF2	3.691	0.951	7	0.763	–	–	–
	Precisely collecting shoring coordinates on-site for meticulous comparison with approved design, ensuring precise execution	10	SRF3	3.569	1.052	9	0.641	–	–	–
	Real-time tracking and monitoring using digital twin's site reconstruction feature to detect and rectify construction deviations and safety issues	9	SRF4	4.01	0.955	1	0.728	–	–	–
	Efficiently managing repair part inventories to ensure timely availability for service technicians before maintenance	13	SRF5	3.79	1.051	5	0.691	–	–	–
	Assess site progress against the schedule and work sequence and respond with clearance or alerts as necessary	10	SRF6	3.975	0.942	2	0.639	–	–	–
	Using wireless sensors on work clothing to locate personnel, determine proximity to construction site hazards, and confirm necessary security measures	11	SRF7	3.469	1.228	10	0.758	–	–	–
	Offering real-time supply chain visibility and automated logistics management, including route adjustments and truck tracking, optimizing transportation efficiency and enhancing on-site safety through IoT	11	SRF8	3.852	1.079	4	0.773	–	–	–
	Installing sensors to manage construction site boundaries by detecting unauthorized door access	11	SRF9	3.667	1.111	8	0.775	–	–	–
	Employing image processing algorithms to evaluate construction site component and material conditions from video or photographic images	9	SRF10	3.951	0.83	3	0.737	–	–	–

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Table 3. Continued

Constructs	Measurement item	Reference	Code	Mean	Standard deviation	Rank	Loadings	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
Office-site-related factors (OSRF)	Continuous real-time enrichment with updated semantic data throughout all construction project phases, including geometric, specifications, drawings, schedules, and financial reports	10	OSRF1	3.963	0.881	2	0.737	0.916	0.93	0.57
	Greater automation complements human workers, allowing them to work independently and adapt to changing circumstances through DT	12	OSRF2	3.605	0.952	9	0.776	–	–	–
	Centralizing project information and artifacts in a single source of truth for enhanced collaboration, constraint removal, and schedule clarity	10	OSRF3	3.728	0.956	4	0.812	–	–	–
	Describing equipment operations and central control unit comparing real-time performance with planned data for cost, quality, speed, and accuracy assessment	15	OSRF4	3.672	1.023	5	0.766	–	–	–
	Real-time construction progress tracking through data integration from various sources into a virtual model, accessible on mobile devices for project managers and analysts	9-13	OSRF5	3.527	0.872	7	0.716	–	–	–
	Collecting real-time on-site progress data for collaboration by comparing it to the project schedule and work sequence	10	OSRF6	4	0.916	1	0.762	–	–	–
	Supporting project management in monitoring work progress, comparing it to plans, and addressing problems for the benefit of stakeholders using human intervention or equipment-based actions	15	OSRF7	3.63	0.999	6	0.727	–	–	–
	Using CPSs and various technologies for real-time data capture and analysis, enabling automated learning, and workflow optimization in construction decision-making	1-2-9	OSRF8	3.568	0.915	10	0.762	–	–	–
	Real-time AI-enabled risk assessment in buildings, diagnosing performance issues and suggesting solutions for implementation in the physical entity	6-14	OSRF9	3.765	0.906	3	0.709	–	–	–
	Providing real-time insights for construction teams with automatic updates of physical assets, revealing their status, working conditions, and positions on the construction site	6	OSRF10	3.617	1.037	8	0.769	–	–	–

Note(s): 1- Kor *et al*. (2023) 2- Yan *et al*. (2022) 3- Hu *et al*. (2021) 4- Voipio *et al*. (2023) 5- Chang *et al*. (2021) 6- Ghansah and Lu (2023) 7- Zhang *et al*. (2022b) 8- Greif *et al*. (2020) 9- Attaran and Celik (2023) 10- Ammar *et al*. (2022) 11- Zhang *et al*. (2022a) 12- Yeung *et al*. (2022) 13- Madubuike *et al*. (2022) 14- Hua *et al*. (2022) 15- Salem and Dragomir (2022)
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on DT usage in both site and office environments, as well as their intersection to provide a comprehensive understanding of how DT technology functions in these settings.

Data collection took place through online methods to capture a wide and diverse set of responses from professionals actively engaged in DT-related construction work, regardless of their geographic location.

3.1 Research methodology

This research employs a hybrid approach that integrates a systematic review with PLS-SEM analysis. The systematic review offers a comprehensive overview of the relevant literature, identifies gaps in the existing body of knowledge, and is supplemented by expert judgments that offer practical, real-world insights and contextual understanding. Meanwhile, PLS-SEM facilitates the quantitative analysis of data that enables the examination of the relationships between the key factors needed for the successful implementation of DT with a focus on three key areas: "DT in Construction Site Environments", "DT in Construction Office Environments", and "DT at the Site-Office Interface". These methods complement each other to create a robust foundation that enhances the validity and depth of the findings of this research, offering rich insights into the factors influencing the successful implementation of DT in construction projects throughout their lifecycle. To systematically address the research purposes, this study followed a structured three-step research approach involving survey design, data collection, and descriptive statistics, followed by detailed data analysis. The primary objective was to conduct a correlational study to reveal relationships among different aspects within a specific context, following Fellows and Liu (2021) approach. Thus, this study used quantitative methods to examine these relationships by constructing a hypothetical model to explore the factors that contribute to the successful implementation of DT in construction projects.

Based on insights from a comprehensive literature review on critical factors in DT application, this study conducted questionnaire survey for data collection due to its wide acceptance, efficiency, and cost-effectiveness, as recommended by Adabre and Chan (2019).

To test the research hypotheses, this study utilized Confirmatory Factor Analysis (CFA) and Structural Equation Modeling (SEM). SEM is a powerful multivariate technique used to estimate interdependent relationships among variables (Yuan *et al*., 2018). It enables users to visually and systematically understand complex relationships through path diagrams (Kim *et al*., 2009). CFA, introduced by Gorsuch (1990), is employed to explicitly test hypotheses regarding factor analytic problems by analyzing observed variables and latent constructs. It assesses how observed variables measure these underlying constructs and models their relationships. SEM/CFA, as highlighted by Molenaar *et al*. (2000), is a statistical method that adeptly demonstrates relationships between research variables. CFA was employed to validate the reliability and appropriateness of factor structures related to successful DT application, and SEM was used to evaluate the causal relationships between successful DT implementation and the discussed factor structures. Additionally, SEM examined interactive effects among these factor structures on DT application in construction projects.

Two primary types of SEM exist: Covariance-Based SEM (CB-SEM) and Partial Least-Squares SEM (PLS-SEM), also known as PLS path modeling (Hair *et al*., 2021). For this study, PLS-SEM was preferred due to its distribution-free nature and suitability for data from unknown distributions (Falk and Miller, 1992). Additionally, PLS-SEM doesn't demand a large sample size (Fornell and Bookstein, 1982; Hair *et al*., 2021). Moreover, employing other methodologies like covariance-based structural equation modeling (CB-SEM) may not yield results for hierarchical models (Hair *et al*., 2012). In a two-stage process, PLS-SEM involves validating the measurement model's reliability and validity in the first stage, followed by assessing the structural model's explanatory power and path coefficients in the second stage (Trinh *et al*., 2019). For instance, (Kineber *et al*., 2023) examined the barriers to implementing Digital Twins in the construction sector, focusing on their impact on

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sustainability. Using a systematic literature review and SEM, the research identified key obstacles to DT adoption and assessed their significance within the sustainability pillars. This analysis also focused on the relation of the barriers of DT implementation with Operational Sustainability Success in order to provide an overview of the way the barriers can be overcome to improve sustainable practices in the construction projects. In this regard, (Alnaser *et al*., 2024) introduced a new innovative framework based on SEM to assess BIM-DT adoption readiness for sustainable construction. By the SEM methodology, the study investigated the interplay between key success factors and sustainable parameters to offer an overview of how KSFs affect the deployment of BIM-DT. Furthermore, (Naji *et al*., 2024) developed an integrated model in order to assess the readiness of buildings regarding digital transformation at the pre-construction phase. Using SEM and CFA, key success factors were identified through expert interviews and literature review. In other words, SEM and CFA were used in this study to determine the relationships among the different components related to the readiness of Digital Transformation-Pre-Construction success factors quantitatively and further analyze their effect on the level of readiness. Notably, several studies in the field of construction management have successfully utilized PLS-SEM, as demonstrated by examples such as (Adabre and Chan, 2021; Munianday *et al*., 2022; Naji *et al*., 2022; Trinh *et al*., 2019).

The choice of PLS-SEM in this study aligned with the need for a method capable of analyzing complex relationships between multiple factors involved in the successful implementation of DT in construction projects. Compared to the CB-SEM method, PLS-SEM provides greater flexibility in exploratory studies with small, non-normal samples owing to its nonparametric nature, limitless enablement in testing hypotheses, and more precise interrelationship analysis, along with validity and reliability assessments (Magno *et al*., 2022). In contrast to EFA and multiple regression, PLS-SEM offers capabilities for both the prediction of latent variables and testing hypotheses (Hair *et al*., 2019). Thus, it holds special potential to analyze the complex interactions between site-related, office-related, and office-site-related factors in DT integration across different lifecycle phases.

Table 4 highlighted how PLS-SEM offers a better balance of theoretical rigor and practical flexibility compared to other methods. It also outlined the limitations of other approaches, reinforcing the rationale for selecting PLS-SEM to address the identified research gaps in the DT application in construction projects.

4. Questionnaire development

In this quantitative study, a survey questionnaire was used to evaluate a proposed hierarchical model within the context of DT in construction projects. To identify crucial factors necessary for the effective utilization of DT, an extensive literature review was conducted, guiding the creation of the initial questionnaire draft. To validate the survey's accuracy, a content validity approach was employed based on three criteria: alignment with existing literature, consistency with prior research, and expert judgment (Alashwal and Abdul-Rahman, 2014).

To enhance the instrument's content validity and reliability, a pilot study was conducted with five experienced academics and experts in the area of study, each possessing at least 10 years of research experience, whose information is presented in Table 5. The pilot study focused on individuals with extensive academic and research backgrounds in DT and BIM, along with significant practical experience in construction projects. The target population also included professionals with over a decade of experience in client, consultant, or contractor roles, ensuring a comprehensive understanding of both the theoretical and practical aspects of BIM and DT implementation. They assessed each questionnaire item using a five-point scale, offering feedback that proved invaluable in refining the questionnaire. Their input led to improvements such as clarifying unclear questions, shortening the questionnaire, and explaining specific terms. These expert recommendations were incorporated into the questionnaire.

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Table 4. Comparison of PLS-SEM with other methodologies

Methodology	Description	Advantages	Disadvantages	Potential impact on results
CB-SEM (covariance-based SEM)	- A parametric SEM method that aims to reproduce the covariance matrix of observed variables	- Suitable for theory testing and validation of established theories - Provides goodness-of-fit indices	- Requires a larger sample size - Assumes multivariate normality, making it less flexible in real-world applications	- Likely to provide more precise parameter estimates but unsuitable given the limited sample size in this study - Strict assumptions about normality could undermine the robustness of the results in a practical construction context
Bayesian SEM	- Uses Bayesian inference to estimate model parameters	- Flexibility in dealing with small samples and complex models - Can incorporate prior knowledge into analysis	- Computationally intensive and requires expertise in Bayesian statistics - Results can be highly sensitive to prior distributions	- May yield nuanced insights but requires careful handling of priors, which could be impractical in a dynamic field like construction
Phased analysis	- A sequential method for assessing different model structures over distinct phases	- - Breaks down model complexity by testing each phase sequentially - Easier to identify misspecifications or weak relationships	- Requires multiple stages of validation, which can be time-consuming	- While useful for examining specific phases, it may overlook significant relationships, reducing the overall comprehensiveness of the analysis
Exploratory factor analysis (EFA)	- Data reduction technique to identify latent constructs	- Suitable for exploring the underlying structure of a large set of variables - Requires fewer assumptions about the data	- Not designed for hypothesis testing or prediction - Focused on exploratory rather than confirmatory analysis	- May reveal factors in DT implementation but would not allow for hypothesis testing or examining relationships, limiting its applicability to this study
Multiple regression analysis	- A method used to predict the value of a dependent variable based on multiple independent variables	- Straightforward for estimating linear relationships between variables - Suitable for hypothesis testing and theory building	- Assumes linear relationships between variables - Not suitable for complex models with latent variables or mediating effects	- Could oversimplify relationships between factors influencing DT implementation, given the complexity of constructs in the construction context

Source(s): Authors' own work

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Table 5. Demographic profiles of the experts

Job title	Occupational category	Educational background	DT/BIM work experience
Project manager	Consultant and Design/Engineering	Master's degree	38
BIM manager	Consultant and Design/Engineering/ Owner	Master's degree	25
BIM modeler	Consultant and Design/Engineering/ University	PHD	17
Architect/BIM manager	Consultant and Design/Engineering/ University	PHD	15
BIM coordinator	Contractor/University	Master's degree	15

Source(s): Authors' own work

The final version of the questionnaire was structured into four sections. The first collected respondents' general information like education, work experience, and employing organization. The subsequent section focused on criteria for assessing the characteristics of DT implementation in construction projects. The remaining three sections of the questionnaire were dedicated to gauging the relative significance of factors pertaining to the implementation of DT in the three key areas previously discussed. Respondents, drawing from their practical experiences on construction projects and their expertise in DT, were asked to express their level of agreement using a five-point Likert scale, ranging from 1 (indicating "strongly disagree") to 5 (indicating "strongly agree") for each statement presented in these sections.

4.1 Research sampling and data collection

This study utilized a purposive sampling methodology for questionnaire distribution. In contrast to random selection in probability sampling, purposive sampling targets specific characteristics of the population relevant to the research. This method is valuable when researchers need insights from individuals with specialized expertise, especially in uncertain topics (Liu *et al*., 2017). This method was employed to select respondents based on their expertise, specifically in the application of DTs in construction projects throughout their lifecycle, in alignment with the study's objectives. Thus, the target population comprised individuals with experience as clients, consultants, or contractors in the construction sector, practical knowledge of BIM and DT implementation in construction projects, and a competent understanding of DT and its application in project management.

One of the key challenges in these studies is the collection of sufficient data to validate the model. CB-SEM is sensitive to data normality and the substantial input data required, unlike PLS-SEM, as suggested by Hair *et al*. (2012), requires fewer data points for model validation, especially concerning the model's dimensions. By a "rule of thumb", the required number of samples is estimated to be at least ten times the maximum value of two criteria: the maximum number of relationships between a latent construct and observed variables and the maximum number of relationships between latent constructs (Ahmadabadi and Heravi, 2019b).

Given the assumptions about relationships between observed variables and latent constructs, the proposed model requires a sample size of 30. While the "10 times" rule is commonly used for minimum sample size calculations (Kock and Hadaya, 2018; Goodhue *et al*., 2012) have questioned its accuracy. (Kock and Hadaya, 2018) confirmed the findings of (Goodhue *et al*., 2012) in a study on the minimum sample size for PLS-SEM. Additionally, (Iacobucci, 2010) recommends 150 respondents for CFA, while (Kline, 2014) suggests a minimum of 100 respondents for CFA. Nevertheless, A sample size ranging from 100 to 200 is also considered adequate for the CFA of SEM (Liu *et al*., 2017).

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In light of these recommendations and in comparison to prior studies in similar domains, for instance (Adabre and Chan, 2021; Ikuabe *et al*., 2022), this study distributed 200 questionnaires using online methods such as Google Forms, LinkedIn, email, and messenger platforms. Participants were provided with an explanation of the research objectives, and their responses were ensured to be confidential and anonymous. A total of 122 completed surveys were received, resulting in a response rate of 61%. These 122 respondents represented a homogeneous sample comprising project managers, project engineers, BIM experts, and academic experts in DT from both Iran and various regions across North America, Europe, Asia-Pacific, and the Middle East. Table 6 in this study provides a breakdown of the respondents' socio-demographic details collected during the data collection process.

5. Results

In this study, the authors employed SmartPLS version 4 to perform confirmatory factor analysis and to assess the statistical support for this research's hypotheses through structural equation modeling, utilizing the partial least squares (PLS) methodology.

5.1 Assessment of the measurement model

In this research, CFA was employed to appraise the correspondence between a construct's measurements and the theoretical concept they represent (Gumusburun Ayalp, 2022). The measurement model's reliability and validity were assessed using various criteria, such as composite reliability, factor loadings, and average variance extracted (AVE) (Munianday *et al*., 2022). Internal reliability was evaluated through composite reliability and Cronbach's alpha, with both values exceeding the 0.7 threshold (Hair *et al*., 2021). Indicator reliability was also scrutinized, with a minimum loading of 0.4 on their corresponding constructs (Trinh *et al*., 2019). Convergent validity was determined by evaluating the AVE, which should exceed 0.5 (Hair *et al*., 2021).

Table 6. Demographic profiles of the respondents

Characteristic	Category	Frequency	Percentage (%)
Gender	Female	34	27.8
	Male	88	72.2
Age	20–25 years	7	5.7
	25–30 years	42	34.4
	30–35 years	44	36.1
	35–40 years	16	13.2
	>40 years	13	10.6
Educational background	PhD	6	4.9
	Master's degree	88	72.1
	Bachelor's degree	28	23
Occupational category	University	13	10.6
	Owner/Client	15	12.3
	Contractor	24	19.7
	Consultant and Design/Engineering Firm	70	57.4
DT/BIM work experience	<1 year	5	4.1
	1–5 years	34	27.9
	5–10 years	28	22.9
	10–15 years	44	36.1
	15–20 years	6	4.9
	>20 years	5	4.1
DT-related knowledge	Novice	0	0
	Intermediate	8	7
	Proficient	65	53
	Expert	49	40

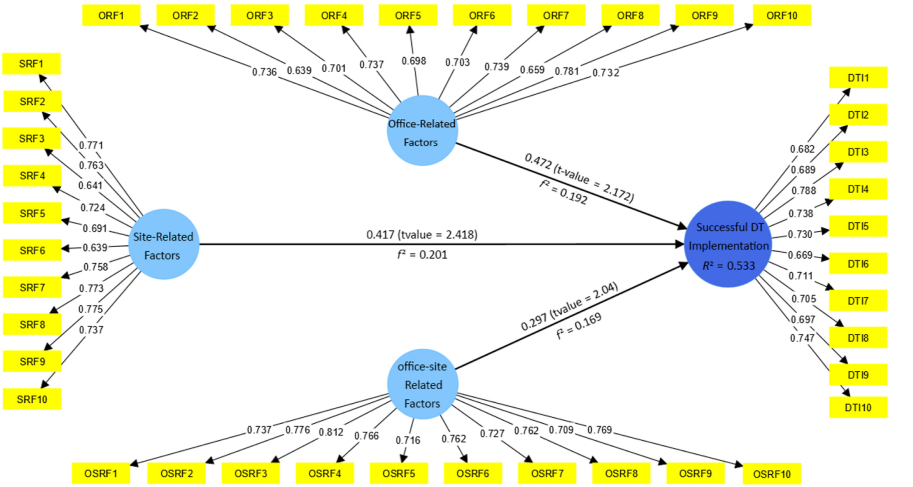
Source(s): Authors' own work

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The assessment results, displayed in Table 3 and Figure 3, demonstrate that all variables exceeded the recommended thresholds of 0.4 and 0.5 for loadings and AVE values, confirming strong convergent validity for indicators and constructs. Furthermore, each construct demonstrates satisfactory levels of internal consistency and reliability, as evidenced by composite reliability and Cronbach's alpha values exceeding the requisite threshold of 0.7 (Munianday *et al*., 2022).

Next, mean values, standard deviations, and rank order based on mean values were computed for both the indicators of successful DT implementation and the factors associated with the packages, as detailed in Table 3. Regarding the indicators, the mean scores span a range from 4.185 to 3.889 across the ten indicators, while for the 30 factors, they varied between 4.049 and 3.469. Notably, all mean values exceeded the threshold of "less important" (<2), demonstrating the respondents' recognition of the significance of all 10 indicators and 30 factors in assessing successful DT implementation. Moreover, most indicators exhibited low standard deviations (<1), reflecting a high level of consensus among the respondents in evaluating these indicators (Adabre and Chan, 2021).

Subsequently, Discriminant validity, which measures how distinct a construct is from others, was evaluated (Hulland, 1999). It requires the square root of each construct's AVE to exceed the inter-construct correlations, and the loading of a measurement item on its respective construct to surpass the cross-loadings (Fornell and Larcker, 1981). The findings of this study (Table 7) showed that the square root of AVEs (highlighted on the diagonal) exceeded



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Figure 3. Model of factor packages effect on the successful DT implementation

Table 7. Comparison of square-rooted AVEs and correlation coefficient between constructs

	Successful DT implementation	Office-related	Site-related	Office-site-related
Successful DT implementation	0.716	–	–	–
Office-related	0.604	0.711	–	–
Site-related	0.623	0.695	0.729	–
Office-site-related	0.646	0.650	0.674	0.755

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correlation coefficients between any pair of latent constructs, displayed in the corresponding rows and columns (off-diagonal values), demonstrating strong discriminant validity among the latent constructs. Additionally, Discriminant validity was confirmed through cross-loading analysis (Table 8), which revealed that each variable had stronger loadings on their intended constructs compared to other constructs in the model, reinforcing the discriminant validity of the constructs (Munianday *et al.*, 2022).

5.2 Assessment of the structural model

During the final stage of model development, bootstrapping with 5,000 samples was employed, as recommended by Hair *et al.* (2021), to assess the path coefficients' significance and validate this study's hypotheses. To confirm the assumptions, t-values for the model's path hypotheses

Table 8. Analysis of cross loadings

Items	Successful DT implementation	Office-related	Site-related	Office-site-related
DTI3	0.788	0.503	0.524	0.54
DTI10	0.747	0.46	0.566	0.522
DTI4	0.738	0.45	0.499	0.469
DTI5	0.73	0.412	0.424	0.464
DTI7	0.711	0.523	0.558	0.545
DTI8	0.705	0.433	0.383	0.466
DTI9	0.697	0.461	0.539	0.536
DTI2	0.689	0.473	0.408	0.474
DTI1	0.682	0.442	0.48	0.493
DTI6	0.669	0.419	0.334	0.407
ORF9	0.48	0.781	0.568	0.62
ORF7	0.422	0.739	0.595	0.561
ORF4	0.463	0.737	0.454	0.554
ORF1	0.545	0.736	0.487	0.598
ORF10	0.415	0.732	0.565	0.599
ORF6	0.466	0.703	0.417	0.485
ORF3	0.498	0.701	0.527	0.541
ORF5	0.384	0.698	0.357	0.485
ORF8	0.455	0.659	0.587	0.574
ORF2	0.453	0.639	0.442	0.413
SRF9	0.487	0.563	0.775	0.482
SRF8	0.442	0.324	0.773	0.518
SRF1	0.517	0.474	0.771	0.567
SRF2	0.532	0.48	0.763	0.62
SRF7	0.462	0.504	0.758	0.59
SRF10	0.34	0.45	0.737	0.609
SRF4	0.52	0.519	0.728	0.557
SRF5	0.507	0.495	0.691	0.521
SRF3	0.487	0.348	0.641	0.429
SRF6	0.498	0.423	0.639	0.428
OSRF3	0.529	0.503	0.585	0.812
OSRF2	0.527	0.509	0.604	0.776
OSRF10	0.5	0.516	0.51	0.769
OSRF4	0.608	0.401	0.508	0.766
OSRF8	0.513	0.533	0.488	0.762
OSRF6	0.463	0.495	0.515	0.762
OSRF1	0.474	0.533	0.516	0.737
OSRF7	0.573	0.471	0.576	0.727
OSRF5	0.503	0.489	0.584	0.716
OSRF9	0.494	0.503	0.425	0.709

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were calculated, with acceptable thresholds set at 2.58 for $p < 0.01$, 1.96 for $p < 0.05$, and 1.65 for $p < 0.10$, following (Ahmadabadi and Heravi, 2019a). The hypotheses were evaluated based on the structural model, where each path represents a hypothesis. The statistical significance of path coefficients was assessed, and the strength of influence between constructs was categorized as weak (0.1–0.3), moderate (0.3–0.5), or strong (0.5–1.0), as suggested by Murari (2015). The SEM analysis (Figure 3) revealed a moderate influence ($\beta^1/_5$ 0.472, $p < 0.05$) of "office-related factors" on "successful DT implementation," a similar level of influence ($\beta^1/_5$ 0.413, $p < 0.05$) from "site-related factors," and a weaker influence ($\beta^1/_5$ 0.297, $p < 0.05$) from "office-site related factors" on "successful DT implementation." Consequently, three hypothesis paths were supported, as their *t*-values exceeded 1.96 (t0.05 > 1.96).

To assess the validity of the structural model, coefficients of determination (R^2) and effect size (f^2) values were computed. The structural model's ability to explain variance is assessed using the coefficient of determination (R^2) [106], which quantifies the proportion of variability in the successful DT implementation construct explained by associated factor constructs (Chin, 1998). R^2 values vary between 0 and 1, with higher values indicating greater predictive and explanatory capabilities of the structural model. In accordance with (Hair *et al*., 2019), R^2 values of 0.75, 0.50, and 0.25 are considered significant, moderate, and weak, respectively.

Besides assessing R^2 values for all endogenous constructs, the f2 metric measures how much the R^2 value changes when specific exogenous constructs are excluded from the model (Hair *et al*., 2021). Construct effect sizes are classified as small ($0.02 \leq f2 < 0.15$), medium ($0.15 \leq f^2 < 0.35$), or large ($f2 \geq 0.35$) (Cohen, 2013). Overall, R^2 and f^2 values validate the structural model and demonstrate its predictive utility for successful DT implementation based on 10 defined criteria. The R^2 value for successful DT implementation is 0.533, indicating that three constructs (office-related, site-related, and office-site related) account for 53% of the variation in successful DT implementation. This reflects a satisfactory level of predictive accuracy and model quality (Hair *et al*., 2014). Figure 3 illustrates that "site-related factors" have a major effect (0.201) on "successful DT implementation", while "office-related factors" and "office-site related factors" have medium effects (0.192, 0.169), respectively.

6. Discussion

This study compiled best practices and critical factors for assessing the influence of the factors on successfully DT implementation in construction projects during their lifecycle. Unlike prior research that focused on specific aspects of DT in construction projects, this research expands understanding of DT utilization in construction projects by exploring the link between office-related factors (ORF), site-related factors (SRF), and office-site related factors (OSRF), revealing three constructs contributing to successful DT implementation. The study outcomes demonstrate that ORF construct exerts a favorable and substantial impact on successful DT implementation.

ORF package, which has 10 indicators, is a significant contributor to measuring successful DT implementation, and had the highest average path coefficient compared with the other constructs. The positive and significant path coefficient value of 0.472 at the 0.05 level underscores the importance of effective management and collaboration within office environments in influencing the success of DT deployment. This finding aligns with (Salem and Dragomir, 2022), who emphasized the role of AI and data analytics in optimizing project management.

The SRF construct, which is also characterized by 10 measurable elements, is a key contributor to assessing the successful implementation of the DT construct. The route coefficient value of 0.413 supports this and highlights the necessity of a comprehensive approach to incorporating site dynamics. This is consistent with (Kosse *et al*., 2022), who discussed DT's capability to foster informed decision-making through real-time data integration. The identified factors within the ORF and SRF constructs resonate with the frameworks established in previous studies, indicating that successful DT implementation requires a synergistic relationship between office and site environments to enhance decision-making processes and project efficiency.

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The construct of OSRF, which has 10 indicators, also makes a significant contribution to successful DT implementation, but less than OSRF and SRF. Notably, while the OSRF construct demonstrates a route coefficient value of 0.297, indicating its contribution to successful DT implementation, this also signals an area requiring further exploration. This aligns with existing literature highlighting gaps in the application of DT within construction office and site environments. Furthermore, findings of this research resonate with (Xie *et al*., 2022), emphasizing the need for ongoing research into the integration of DT applications across various lifecycle phases, an area that remains underexplored in current literature.

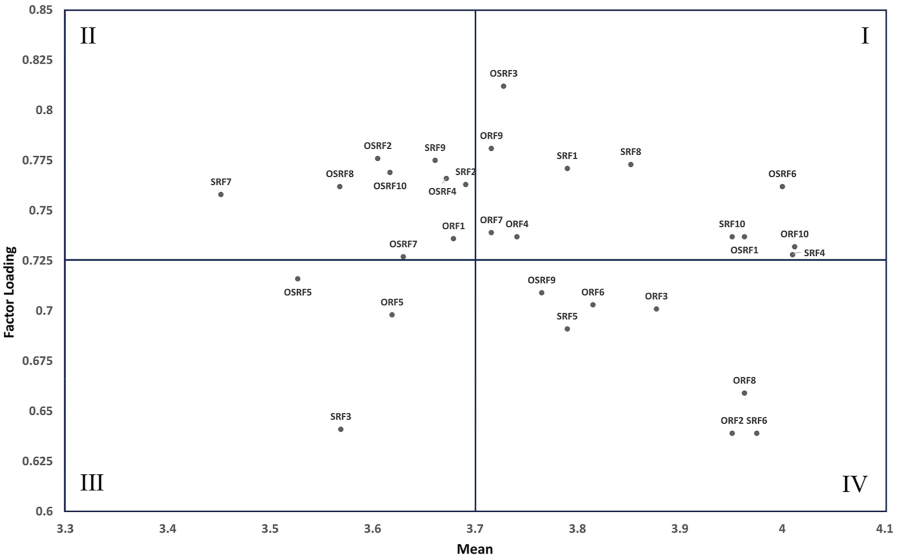
The results of this study align with the findings of Ghansah and Lu (2023), highlighting the importance of DT implementation in both office and site environments for achieving effective building performance, process efficiency, improved information management, and enhanced user interactions and suggest that DTs offers a critical analytic advantage by aggregating data from various sources across construction offices and sites. The research also agrees with a study done by Hu *et al*. (2022), who indicate that in light of growing automation in the construction industry, the DTs of the construction sites have cost-effective solutions for real-time multi-asset connectivity, simulations, and decision support using Industry 4.0 technologies. However, there remains a lack of comprehensive reviews that explore the influencing factors for successful DT implementation across the construction project lifecycle.

To enhance the impact of results, it is essential to offer detailed insights into the intricate relationships among the identified factors for successful DT implementation In Table 3, mean values and CFA were utilized to assess the relative significance and contributions of factor packages in successful DT implementation. While mean values and loading coefficients provide insight into a factor's importance, they may not always align in terms of ranking. In some cases, an indicator with a high mean value may not exhibit correspondingly high loading coefficients (Chou *et al*., 2012). Hence, it is advisable to combine mean values and loading coefficients to comprehensively assess key factors within the three packages (Yuan *et al*., 2018). To visually represent this assessment of each indicator calculated from mean value and CFA (loading coefficients), a two-dimensional significance analysis was conducted, plotting mean values on the *x*-axis and loading coefficients on the *y*-axis in Figure 4. Indicators that are considered more important are positioned in Quadrant I, where both their mean values and loading coefficients exceed the average. Conversely, indicators in Quadrant III are deemed less important, as both mean values and loading coefficients fall below the average. Indicators that have either the mean value or loading coefficient surpass the average are labeled as important and can be found in Quadrant IV or II (Figure 4).

The 2D significance cross-analysis (see Figure 4) revealed that eleven indicators resided in Quadrant I, signifying their heightened importance in ensuring successful DT implementation. These crucial factors will be introduced and examined in the subsequent discussion.

(1) Key factors from ORF package (office-related): ORF4, ORF7, ORF9, ORF10. During the digital twin design stage in the construction office, the extraction and analysis of end-user requirements deeply align with project goals, ensuring a tailored approach that adapts to stakeholders' needs. Integrating real-time AI capabilities into the office ecosystem transforms decision-making by leveraging the DT database, potentially automating processes and enhancing overall project efficiency. This proactive approach streamlines decision-making and offers actionable options for managers. Simultaneously, streamlining project status monitoring ensures compliance with contract documents and industry norms, facilitated by automated inventory management. Considering these factors is crucial for successful DT implementation in construction, enabling swift, informed decision-making or any necessary deviations or adjustments, fostering a highly efficient and adaptable office environment.

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Figure 4. Two-dimensional importance analysis diagram of key factors

(2) Key factors from SRF package (Site-related): SRF1, SRF4, SRF8, SRF10. In construction projects, successful DT implementation relies on a holistic approach that incorporates various site-related factors. Sensing devices are strategically placed across the site to collect real-time data on space, boundaries, roads, facilities, equipment, materials, and personnel. This data stream provides a comprehensive view of the site, enabling real-time tracking, monitoring, and quick resolution of construction deviations and safety issues. Simultaneously, DT ensures real-time supply chain visibility, automated logistics management, including route adjustments, truck tracking, and transportation efficiency optimization, all while bolstering on-site safety through IoT integration. Image processing algorithms further enhance insight by evaluating site components and materials using visual data. This integrated approach guarantees efficiency, safety, and project integrity, making the four key factors in the SRF package pivotal for successful DT implementation in construction projects.

(3) Key factors from SRF package (Site-related): OSRF1, OSRF3, OSRF6. To successfully implement DT technology in construction projects, it is imperative to consider various critical factor at the construction site-office interface. Continuous updates of real-time semantic data, including geometry, schedules, and financial reports, are essential for well-informed decision-making. This seamless data exchange between the field and the office ensures timely information, aligning on-site actions with project goals. Real-time on-site progress data analysis in comparison to planned schedules is vital for proactive issue resolution, ensuring efficient project management and project alignment. Furthermore, the centralized project information repository acts as a crucial link between the physical site and administrative functions, enhancing collaboration, reducing errors, and promoting clear communication. These three key factors in the SRF package significantly drive successful DT implementation, forming a cohesive foundation that enhances construction project success.

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7. Implication

7.1 Managerial implications

The findings from this study provide practical guidelines for construction managers and professionals in the field who seek to implement the DT technology effectively at various stages of construction projects. Managers are encouraged to prioritize the integration of real-time data platforms and AI-based decision-making systems to drive dynamic communication and proactive monitoring. These steps greatly help in enhancing operational efficiency by cutting delays in approvals or decision-making processes. Additionally, investment in sensor-based monitoring tools and field data integration mechanisms is important for ensuring streamlined and error-free site operations. These tools contribute to improved safety metrics and aligning the site activities with project timelines.

To address the critical office-site interface, hybrid workflows need to be developed that can bridge this gap by incorporating digital data into the traditional practices. Establishing collaborative protocols can minimize miscommunication and therefore reduce costly errors during handovers or phased execution. Together, these recommendations make a comprehensive guide for managers on ways to minimize material wastage, enhance predictive accuracy, and make stakeholder expectations match the measurable project outcomes.

7.2 Theoretical implications

This research takes an important step toward advancing the understanding of DT technology in the construction industry by addressing critical gaps in current knowledge. One of its main contributions is the development of an integrated framework that ties together office-related, site-related, and office-site-related factors. This framework not only highlights the dynamics of DT implementation but also provides a structured approach for exploring how these factors interact and influence outcomes. Unlike earlier studies, it takes a closer look at the hierarchy of these influences, offering fresh insights and a new perspective for analysis.

In addition to its conceptual contributions, the study uses advanced analytical methods like PLS-SEM and CFA to validate the proposed framework. These tools make it possible to explore the complex relationships between the factors, setting a high standard for future research in this area. By concentrating specifically on the construction sector, the research also provides targeted insights that address unique industry challenges, such as managing on-site unpredictability and ensuring smooth coordination between offices and sites. This practical focus helps bridge broader gaps in the general literature on DT, anchoring the findings in real-world scenarios and making them more applicable to industry professionals.

8. Conclusion and limitation

The notion of digital twins has the capacity to enhance the efficiency and productivity of construction projects across their lifecycle by enabling data-driven decision-making through real-time monitoring, predictive analytics and streamlining of processes. This study investigates the essential factors for the effective implementation of DT for control and monitoring in different stages of a construction project's life cycle. It specifically concentrates on three areas: the office, the on-site, and the transitional zone connecting the office and the site. By comprehensively reviewing existing research, we've identified the crucial factors affecting DT integration.

A questionnaire survey of 122 respondents was conducted to assess the significance of various factors. Data analysis involved using descriptive methods and PLS-SEM to establish a model examining the relationship between successful DT implementation and three categories of factors. The CFA testing confirmed a good fit between the hypothesized model and survey data, suggesting that all indicators contributed to their respective packages and ultimately to the successful DT implementation. The study's findings highlighted significant relationships among factor constructs and successful DT implementation, with office-related factors having

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a notably positive influence. Site-related factors also made substantial contributions, while office-site related factors had a somewhat less pronounced impact. Additionally, the study identified eleven key factors based on survey means and CFA factor loadings, offering valuable insights for construction professionals aiming for successful DT implementation.

In summary, these implemented key factors will revolutionize DT in construction by improving performance, safety, and efficiency throughout a project's life cycle. However, it's essential to acknowledge certain limitations in this research, such as the relatively small sample size in comparison to the broader target community, restricted access to a wider array of experts within the industry, and the constraints associated with previous studies that could impact the precision and comprehensiveness of the factors considered.

Despite these limitations, the study's insights lay a foundation for understanding DT's potential in construction projects, signaling the need for future research to address these constraints and enhance understanding.

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Corresponding author
Esmatullah Noorzai can be contacted at: Esmatullah.Noorzai1980@gmail.com

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